

This paper covers Differential Geometry, Barrett O'Neill,
with the sub Analysis on Manifolds, Munkres.

Chapter 1. Calculus on Euclidean Space.

Def. A real-valued $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ is called smooth if:

$$\frac{\partial^k f}{\partial x_1^{k_1} \partial x_2^{k_2} \partial x_3^{k_3}} \text{ exists and continuous for all } k_1, k_2, k_3 \geq 0.$$

And, given two smooth functions f and g , define

$$(f+g)(p) = f(p) + g(p) \quad \text{and} \quad (fg)(p) = f(p) \cdot g(p).$$

Def. A tuple $V_p = (p, v) \in \mathbb{R}^3 \times \mathbb{R}^3$ is called a tangent vector,

consisting of the point part p and vector part v .

And, the set $T_p(\mathbb{R}^3) \stackrel{\text{def}}{=} \{(p, v) \mid v \in \mathbb{R}^3\}$ is called the tangent space of $\mathbb{R}^3$ at p .

Def. A mapping $V: \mathbb{R}^3 \rightarrow \bigcup_{p \in \mathbb{R}^3} T_p(\mathbb{R}^3) : p \mapsto V_p$ is called vector field on $\mathbb{R}^3$.

Given two vector field V and W , and real-valued f on $\mathbb{R}^3$, define.

$$(V+W)(p) = V(p) + W(p) \quad \text{and} \quad (fV)(p) = f(p)V(p).$$

In particular, the vector fields on $\mathbb{R}^3$ define by

$$U_1(p) = (1, 0, 0)_p \quad \text{and} \quad U_2(p) = (0, 1, 0)_p \quad \text{and} \quad U_3(p) = (0, 0, 1)_p \quad \text{for each } p \in \mathbb{R}^3$$

are called the natural frame field on $\mathbb{R}^3$.

Lemma. Given a vector field V on $\mathbb{R}^3$, there are unique real-valued V_1, V_2, V_3 on $\mathbb{R}^3$

$$\text{s.t.} \quad V = V_1 U_1 + V_2 U_2 + V_3 U_3.$$

Particularly, the functions V_1, V_2, V_3 are called the Euclidean coordinate functions of V .

Proof. By definition, for each point $p \in \mathbb{R}^3$, the vector part of $V(p)$ depends on p , so, denote the vector part of $V(p)$ as $(V_1(p), V_2(p), V_3(p))$. Now,

$$V(p) = (V_1(p), V_2(p), V_3(p))_p = V_1(p) U_1(p) + V_2(p) U_2(p) + V_3(p) U_3(p).$$

And, the uniqueness is given by the comparing the entries of tuples.

Def. Let f be a smooth real-valued function on $\mathbb{R}^3$, and V_p be a tangent vector. The limit $V_p[f] \stackrel{\text{def}}{=} \left. \frac{d}{dt}(f(p + tV)) \right|_{t=0} \left(= \lim_{t \rightarrow 0} \frac{f(p + tV) - f(p)}{t} \right)$

is called the directional derivative of f with respect to V_p .

Lemma. Given a tangent vector $V_p = (V_1, V_2, V_3)_p$ on $\mathbb{R}^3$ and smooth function f ,

$$V_p[f] = \sum_{i=1}^3 V_i \cdot \frac{\partial f}{\partial x_i}(p)$$

Proof. Let $P = (p_1, p_2, p_3) \in \mathbb{R}^3$ be given. Then

$$p + tV = (p_1 + tV_1, p_2 + tV_2, p_3 + tV_3)$$

So that $f(p + tV) = f(p_1 + tV_1, p_2 + tV_2, p_3 + tV_3)$

Now, by the chain rule and the fact that $\frac{d}{dt}(p_i + tV_i) = V_i$,

$$V_p[f] = \left. \frac{d}{dt}(f(p + tV)) \right|_{t=0} = \sum_{i=1}^3 \frac{\partial f}{\partial x_i}(p) V_i. \quad \square$$

Note: In other words,

$$\frac{df}{dt} = \frac{\partial f}{\partial x_1} \cdot \frac{dx_1}{dt} + \frac{\partial f}{\partial x_2} \cdot \frac{dx_2}{dt} + \frac{\partial f}{\partial x_3} \cdot \frac{dx_3}{dt}$$

Thm. Let f and g be functions on $\mathbb{R}^3$, V_p and W_p be tangent vectors, and $a, b \in \mathbb{R}$.

$$1) (aV_p + bW_p)[f] = aV_p[f] + bW_p[f]$$

$$2) V_p[af + bg] = aV_p[f] + bV_p[g]$$

$$3) V_p[f g] = V_p[f] \cdot g(p) + f(p) \cdot V_p[g].$$

Proof. These are obtained by the Lemma above.

Corollary. Given two vector fields V and W on $\mathbb{R}^3$ and real-valued f, g, h on $\mathbb{R}^3$,

$$1) (fV + gW)[h] = fV[h] + gW[h]$$

$$2) V[af + bg] = aV[f] + bV[g] \quad \text{for all } a, b \in \mathbb{R}.$$

$$3) V[f g] = V[f] \cdot g + f V[g].$$

Proof.

$$(a v_p + b w_p)[f] = \sum_{i=1}^{\text{Lemma 3}} (a v_i + b w_i) \cdot \frac{\partial f}{\partial x_i}(p) = a \sum v_i \frac{\partial f}{\partial x_i}(p) + b \sum w_i \frac{\partial f}{\partial x_i}(p).$$

Sec 1.4 Curves in $\mathbb{R}^3$. $[I]$ is any open interval.

Def. A differentiable function $\alpha: I \rightarrow \mathbb{R}^3$ is called a Curve.

Moreover, for $\alpha(t) = (\alpha_1(t), \alpha_2(t), \alpha_3(t))$, the tangent vector

$$\alpha'(t) = \left(\frac{d\alpha_1}{dt}(t), \frac{d\alpha_2}{dt}(t), \frac{d\alpha_3}{dt}(t) \right)_{\alpha(t)}$$

is called the velocity vector of α at t .

Examples for Curves and the velocity vector.

The parametrization

Def. Let $\alpha: I \rightarrow \mathbb{R}^3$ be a curve defined on an open interval I .

If $h: J \rightarrow I$ is a differentiable function on an open interval J , then the composite $\beta = \alpha \circ h: J \rightarrow \mathbb{R}^3$ is a curve called a reparametrization of α by h .

Lemma. If β is the reparametrization of α by h , then the velocity is

$$\beta'(s) = \frac{dh}{ds}(s) \cdot \alpha'(h(s))$$

Proof. Denote $\alpha = (\alpha_1, \alpha_2, \alpha_3)$. Then, by the definition,

$$\beta(s) = (\alpha \circ h)(s) = (\alpha_1(h(s)), \alpha_2(h(s)), \alpha_3(h(s)))$$

Now, by the definition of velocity and the chain rule,

$$\beta'(s) = (\alpha_1'(h(s)) \cdot h'(s), \alpha_2'(h(s)) \cdot h'(s), \alpha_3'(h(s)) \cdot h'(s))$$

$$= h'(s) \cdot (\alpha_1'(h(s)), \alpha_2'(h(s)), \alpha_3'(h(s)))$$

$$= h'(s) \cdot \alpha'(h(s)).$$

Lemma. Let α be a curve in $\mathbb{R}^3$ and let f be a differentiable function on $\mathbb{R}^3$.

Then $\alpha'(t)[f] = \frac{d(f \circ \alpha)}{dt}(t)$

Proof.

Π -form.

Def. A map $\varphi: \bigcup_{p \in \mathbb{R}^3} T_p(\mathbb{R}^3) \rightarrow \mathbb{R}$ is called a 1-form on $\mathbb{R}^3$ if;

$$\varphi(av_p + bw_p) = a \cdot \varphi(v_p) + b \cdot \varphi(w_p) \text{ for all } a, b \in \mathbb{R} \text{ and } v_p, w_p \in T_p(\mathbb{R}^3), \text{ for each } p.$$

And, for a fixed $p \in \mathbb{R}^3$, the restriction $\varphi_p: T_p(\mathbb{R}^3) \rightarrow \mathbb{R}: v_p \mapsto \varphi(v_p)$ is a linear functional.

Moreover, given two 1-forms φ_1, φ_2 and $f, g: \mathbb{R}^3 \rightarrow \mathbb{R}$, define

$$(\varphi_1 + \varphi_2)(v_p) \stackrel{\text{def}}{=} \varphi_1(v_p) + \varphi_2(v_p) \text{ and } (f\varphi)(v_p) = f(p) \cdot \varphi(v_p), \text{ for each } v_p \in \bigcup_{p \in \mathbb{R}^3} T_p(\mathbb{R}^3)$$

And given two vector fields V, W ,

$$\varphi(fV + gW) \stackrel{\text{def}}{=} f\varphi(V) + g\varphi(W)$$

$$\text{and } (f\varphi_1 + g\varphi_2)(V) \stackrel{\text{def}}{=} f\varphi_1(V) + g\varphi_2(V).$$

Def. If $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ is differentiable, the 1-form

$$df: \bigcup_{p \in \mathbb{R}^3} T_p(\mathbb{R}^3) \rightarrow \mathbb{R}: v_p \mapsto v_p[f]$$

is called the differential of f .

Example. Consider the projections $\pi_i: \mathbb{R}^3 \rightarrow \mathbb{R}: (x_i) \mapsto x_i$ ($i=1,2,3$). Then,

$$d\pi_i(v_p) = v_p[\pi_i] = \sum_{j=1}^3 v_j \cdot \frac{\partial \pi_i}{\partial x_j}(p) = v_i.$$

Lemma. Let φ be a 1-form on $\mathbb{R}^3$. Then,

$$\varphi = \sum f_i dx_i \text{ where } f_i = \varphi(U_i)$$

Proof. For fixed $p \in \mathbb{R}^3$, $\varphi_p = \sum \varphi_p \circ U_p \cdot (dx_i)_p$.

Note: for fixed $p \in \mathbb{R}^3$, the tangent space $T_p(\mathbb{R}^3)$ is isomorphic to $\mathbb{R}^3$.

So, the dual sp. $(T_p(\mathbb{R}^3))^*$ is a set of 1-forms at a . Moreover,

the triple $\{(dx_1)_p, (dx_2)_p, (dx_3)_p\}$ is a dual basis for $(T_p(\mathbb{R}^3))^*$, corresponding to $\{U_1(p), U_2(p), U_3(p)\}$.

Chapter 2. Frame field.

Def. Given tangent vectors V_p, W_p at $p \in \mathbb{R}^3$, a tangent vector

$$V_p \times W_p \stackrel{\text{def}}{=} \begin{vmatrix} U_1(p) & U_2(p) & U_3(p) \\ V_1 & V_2 & V_3 \\ W_1 & W_2 & W_3 \end{vmatrix} \stackrel{(*)}{=} \begin{aligned} &U_1(p)(V_2 W_3 - V_3 W_2) \\ &- U_2(p)(V_1 W_3 - V_3 W_1) \\ &+ U_3(p)(V_1 W_2 - V_2 W_1). \end{aligned}$$

is called the cross product of V_p and W_p .

Lemma. Given V_p and W_p in $T_p(\mathbb{R}^3)$, $V_p \times W_p$ is orthogonal to both V_p and W_p ,

and
$$\|V_p \times W_p\|^2 = (V_p \cdot V_p) \cdot (W_p \cdot W_p) - (V_p \cdot W_p)^2.$$

In other say, $\|V \times W\| = \|V\| \cdot \|W\| \cdot \sin \theta$

Proof. Since U_1, U_2, U_3 are orthonormal,

$$V \cdot (V \times W) = \begin{vmatrix} V_1 & V_2 & V_3 \\ V_1 & V_2 & V_3 \\ W_1 & W_2 & W_3 \end{vmatrix} = 0 \quad \text{and} \quad W \cdot (V \times W) = \begin{vmatrix} W_1 & W_2 & W_3 \\ V_1 & V_2 & V_3 \\ W_1 & W_2 & W_3 \end{vmatrix} = 0.$$

Therefore, the orthogonality is proved. And, to show the second assertion,

$$(V \cdot V) \cdot (W \cdot W) - (V \cdot W)^2 = (V_1^2 + V_2^2 + V_3^2)(W_1^2 + W_2^2 + W_3^2) - (V_1 W_1 + V_2 W_2 + V_3 W_3)^2$$

$$= \sum_{1 \leq i, j \leq 3} V_i^2 \cdot W_j^2 - \left(\sum_{1 \leq i \leq j \leq 3} V_i W_i V_j W_j \right)$$

$$= \sum_{1 \leq i, j \leq 3} V_i^2 \cdot W_j^2 - \left(\sum_{i=1}^3 V_i^2 W_i^2 + 2 \cdot \sum_{1 \leq i < j \leq 3} V_i W_i V_j W_j \right)$$

$$= \sum_{1 \leq i < j \leq 3} V_i^2 \cdot W_j^2 - 2 \cdot \sum_{1 \leq i < j \leq 3} V_i W_i V_j W_j$$

Meanwhile,

$$\|V \times W\|^2 = (V_2 W_3 - V_3 W_2)^2 + (V_1 W_3 - V_3 W_1)^2 + (V_1 W_2 - V_2 W_1)^2$$

$$= V_2^2 W_3^2 + V_3^2 W_2^2 - 2 V_2 W_2 V_3 W_3$$

$$+ V_1^2 W_3^2 + V_3^2 W_1^2 - 2 V_1 W_1 V_3 W_3$$

$$+ V_1^2 W_2^2 + V_2^2 W_1^2 - 2 V_1 W_1 V_2 W_2$$

These are same. $\square$

Sec 2.2 Curves.

Recall: a differentiable function $\alpha: I \rightarrow \mathbb{R}^3$ is called a Curve on an open interval I .

Def. Given a Curve $\alpha: I \rightarrow \mathbb{R}^3$,

$$\alpha'(t) \stackrel{\text{def}}{=} \left(\frac{d\alpha_1}{dt}(t), \frac{d\alpha_2}{dt}(t), \frac{d\alpha_3}{dt}(t) \right)_{d\alpha}$$

is called the velocity of α , and $\|\alpha'(t)\|$ is called the speed of α .

And, for $[a, b] \subset I$, $\int_a^b \|\alpha'(t)\| dt$

is called the arc length of α from $t=a$ to $t=b$.

Def. A curve $\alpha: I \rightarrow \mathbb{R}^3$ is called regular if:

$$\alpha'(t) \neq 0 \text{ for all } t \in I.$$

Thm. Given a regular Curve α in $\mathbb{R}^3$,

there exists a reparametrization β of α such that $\|\beta'(s)\| = 1$ for all s .

Proof. Let $\alpha: I \rightarrow \mathbb{R}^3$ be a regular Curve. Let $a \in I$ be fixed.

Define $S(t) = \int_a^t \|\alpha'(u)\| du$. Since $\|\alpha'\| \neq 0$, the $S(t)$ is monotone increasing on I .

Therefore, there is an inverse function $t: J \rightarrow I$ s.t. $S(t(x)) = x$ for all $x \in J$,

where $J = S[I]$. Now, for $\beta = \alpha \circ t$,

$$\beta'(s) = \frac{dt}{ds}(s) \cdot \alpha'(t(s)) \text{ and } \|\beta'(s)\| = \underbrace{\frac{dt}{ds}(s)}_{>0} \|\alpha'(t(s))\| = \frac{dt}{ds}(s) \cdot \frac{ds}{dt}(t(s)) = 1. \quad \square$$

Def. Given a curve $\alpha: I \rightarrow \mathbb{R}^3$, the function

$$\gamma: I \rightarrow \bigcup_{p \in \mathbb{R}^3} T_p(\mathbb{R}^3); t \mapsto V_{\alpha(t)}$$

is called a vector field on Curve α .

Sec 2.3 Frenet Formula.

Def. Let $\beta: I \rightarrow \mathbb{R}^3$ be a unit-speed curve.

A vector field β' is called the unit tangent vector field on β , denote $T = \beta'$.

The derivative $T' = \beta''$ is called the Curvature vector field of β .

The value $\kappa(s) = \|T'(s)\|$ ($s \in I$) is called the Curvature function of β .

To restriction s.t. $\kappa > 0$,
the unit-vector field $N = \frac{T'}{\kappa}$ is called the principal normal vector field of β .

The cross product $B = T \times N$ on β is called the binormal vector field of β .

Lemma. Let β be a unit-speed curve in $\mathbb{R}^3$ with $\kappa > 0$,

Then, the three vector fields T, N , and B on β are unit vector fields that are mutually orthogonal at each point.

Proof. By definition, $\|T\| = \|\beta'\| = 1$. And since the curvature $\kappa > 0$, $\|N\| = \frac{1}{\kappa} \cdot \|T'\| = 1$.

And, since $\|T\|^2 = T \cdot T$ is constant, $2T \cdot T' = 0$, so $T \perp N$.

therefore the binormal $\|B\| = \|T \times N\| = \|T\| \cdot \|N\| = 1 \cdot 1 = 1$.

In summary, $T = \beta'$, $N = T'/\kappa$, $B = T \times N$, and $\{T, N, B\}$ is orthonormal.

Since $B \cdot T = 0$, $B' \cdot T + B \cdot T' = 0$ by differentiate. Therefore,

$$B' \cdot T = -B \cdot T' = -B \cdot \kappa N = 0$$

And since B is unit, $B' \cdot B = 0$, thus, there is a scalar value function τ s.t.

$$B' = -\tau N$$

, called the torsion. Now

Thm. Let β be a unit-speed curve in $\mathbb{R}^3$ with curvature $\kappa > 0$ and torsion τ ,

$$\begin{aligned} T' &= \kappa N \\ N' &= -\kappa T + \tau B \\ B' &= -\tau N \end{aligned} \quad \text{or} \quad \begin{bmatrix} T' \\ N' \\ B' \end{bmatrix} = \begin{bmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{bmatrix} \begin{bmatrix} T \\ N \\ B \end{bmatrix}$$

Proof.

Curve.

Def. A differentiable function $\alpha: (a, b) \rightarrow \mathbb{R}^3$ is called a Curve.

And the tangent vector

$$\alpha'(t) \stackrel{\text{def}}{=} \left(\frac{d\alpha_1}{dt}(t), \frac{d\alpha_2}{dt}(t), \frac{d\alpha_3}{dt}(t) \right) \stackrel{\text{def}}{=} \sum_{i=1}^3 \frac{d\alpha_i}{dt}(t) \cdot e_i(dt).$$

is called the velocity vector of α at t .

If $h: (c, d) \rightarrow (a, b)$ is a differentiable, then the composition

$$\beta = \alpha \circ h: J \rightarrow \mathbb{R}^3$$

is called a reparametrization of α by h .

Lemmu. If β is the reparametrization of α by h , then

$$\beta'(s) = h'(s) \cdot \alpha'(h(s)).$$



Chapter 3. Euclidean Geometry.

Preliminary: Mappings.

Def. Given a function $F: \mathbb{R}^n \rightarrow \mathbb{R}^m$, define $f_i = \pi_i \circ F$ for each $1 \leq i \leq m$. Then

$$F(p) = (f_1(p), \dots, f_m(p)) \text{ for all } p \in \mathbb{R}^n$$

these $f_1, \dots, f_m$ are called the Euclidean coordinate functions of F .

If the coordinate functions are differentiable, then F is a mapping from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Def. Given a Curve $\alpha: I \rightarrow \mathbb{R}^n$ and mapping $F: \mathbb{R}^n \rightarrow \mathbb{R}^m$, a Composition

$$\beta: F \circ \alpha: I \rightarrow \mathbb{R}^m$$

is called the image of α under F .

Def. Let $F: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a mapping define

$$F_*: \bigcup_{p \in \mathbb{R}^n} T_p \mathbb{R}^n \rightarrow \bigcup_{p \in \mathbb{R}^n} T_{F(p)} \mathbb{R}^m: V_p \mapsto \beta'(0)$$

where $\beta: (-\varepsilon, \varepsilon) \rightarrow \mathbb{R}^m: t \mapsto F(p + tV)$.

Prop. Let $F = (f_1, \dots, f_m)$ be a mapping from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Given a tangent vector $V_p \in T_p(\mathbb{R}^n)$,

$$F_*(V_p) = (V_p[f_1], \dots, V_p[f_m])_{F(p)}$$

Proof. Since $\beta(t) = (f_1(p + tV), \dots, f_m(p + tV))$,

$$\left. \frac{d}{dt} f_i(p + tV) \right|_{t=0} = V_p[f_i], \text{ so prove directly by the definition. } \square$$

Note: for fixed $p \in \mathbb{R}^n$, $F_{*p}: T_p(\mathbb{R}^n) \rightarrow T_{F(p)}(\mathbb{R}^m)$ is called the tangent map of F at p . Moreover, the F_{*p} is linear map.

Sec 3.2. The tangent map of an Isometry.

Thm. Let F be an isometry of $\mathbb{R}^3$ with decomposition $F = TC$. Then

$$F_*(V_p) = (C \cdot V)_{F(p)}$$

Chapter 4. Calculus on a surface.

Def. Let $D \subseteq \mathbb{R}^2$ be an open set.

A one-to-one regular mapping $X: D \rightarrow \mathbb{R}^3$ is called a Coordinate Patch.
 X_{*p} is 1-1 for each p .

A subset $M \subseteq \mathbb{R}^3$ is called a Surface if:

for each $p \in M$, there exists a patch X such that $N \subseteq \text{Im } X$,

where N is a neighborhood of p in M .

Example. Let $S^2 = \{(p_1, p_2, p_3) \in \mathbb{R}^3 \mid \sqrt{p_1^2 + p_2^2 + p_3^2} = 1\}$ be given.

Find a proper patch in S^2 covering a neighborhood of the North Pole $(0, 0, 1)$.

Let $D = \{(u, v) \mid u^2 + v^2 < 1\}$, and define

$$X: D \rightarrow \mathbb{R}^3: (u, v) \mapsto (u, v, \underbrace{\sqrt{1-u^2-v^2}}_f)$$

To show this X satisfies the Patch Condition, calculate the Jacobian

$$J(X) = \begin{pmatrix} \frac{\partial u}{\partial u} & \frac{\partial u}{\partial v} \\ \frac{\partial v}{\partial u} & \frac{\partial v}{\partial v} \\ \frac{\partial f}{\partial u} & \frac{\partial f}{\partial v} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -\frac{u}{f} & -\frac{v}{f} \end{pmatrix}$$

Trivially $\text{rank } J = 2$. So, by the inverse function theorem, X_{*p} is locally one-to-one.

And, $X^{-1}: X[D] \rightarrow D: (p_1, p_2, p_3) \mapsto (p_1, p_2)$ is an inverse of X , so one-to-one.

In special case: Given differentiable $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $M = \{(x, y, f(x, y)) \mid x, y \in \mathbb{R}^2\}$ is a surface.

Via the Monge patch $X(u, v) = (u, v, f(u, v))$.

Patch Computation.

Def. Given a patch $X: D \rightarrow \mathbb{R}^3$, for fixed $(u_0, v_0) \in D$, define

$$u \mapsto X(u, v_0) \quad \text{and} \quad v \mapsto X(u_0, v)$$

are called u, v - Parameter Curves, respectively.

And, partial velocities of X at (u_0, v_0) as

$$X_u(u_0, v_0) \stackrel{\text{def}}{=} \left(\frac{\partial X}{\partial u}(u_0, v_0) \right)_{(u_0, v_0)} \quad \text{and} \quad X_v(u_0, v_0) \stackrel{\text{def}}{=} \left(\frac{\partial X}{\partial v}(u_0, v_0) \right)_{(u_0, v_0)}$$

Def. A regular mapping $X: D \rightarrow \mathbb{R}^3$ s.t. $X[D] \subseteq M$ for the surface M is called a parametrization of the region $X[D]$ in M .

Thm. Given a mapping $X: D \rightarrow \mathbb{R}^3$ where $D \subseteq \mathbb{R}^2$ is open,

X is regular if and only if the cross product $X_u \times X_v \neq 0$ for all $p \in D$.

Proof. By definition, X is regular if and only if given $p \in D$, the Jacobian

$DX(p)$ is one-to-one, that is, $\ker DX(p) = \{0\}$, or $\text{rank } DX(p) = 2$.

The geographical patch in the sphere.

$$\text{Let } \Sigma \stackrel{\text{def}}{=} \left\{ (x, y, z) \in \mathbb{R}^3 \mid \sqrt{x^2 + y^2 + z^2} = r \right\} \text{ where } r > 0.$$

Define a patch

$$X: (-\pi, \pi) \times \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) : (u, v) \mapsto (r \cdot \cos u \cdot \cos v, r \cdot \sin u \cdot \cos v, r \cdot \sin v)$$

$$= r \cdot \cos v \cdot (\cos u, \sin u, 0) + r \cdot \sin v \cdot (0, 0, 1),$$

$$= r \cdot \cos v \cdot \cos u \cdot e_1 + r \cdot \cos v \cdot \sin u \cdot e_2 + r \cdot \sin v \cdot e_3.$$

Calculate the partial velocities

$$X_u(u, v) = (-r \cdot \sin u \cdot \cos v, r \cdot \cos u \cdot \cos v, 0)$$

$$X_v(u, v) = (-r \cdot \cos u \cdot \sin v, -r \cdot \sin u \cdot \sin v, r \cdot \cos v).$$

Surface Criterion.

Let $g: \mathbb{R}^3 \rightarrow \mathbb{R}$ be differentiable, and $c \in \mathbb{R}$, and let $M = \{(x, y, z) \in \mathbb{R}^3 \mid g(x, y, z) = c\}$.
If the differential dg is not zero for all $p \in M$, then M is a surface.